

Activity Dossier

Lab 2 — Predictive Maintenance

AI & Data Science with GenAI (Rail) · Your first machine-learning lab
Student package — everything you need to do this activity

Field	Detail
Module	Statistics / Machine Learning foundations
Builds on	Lab 1 (pandas + plotting) and your classification / model-validation lectures
Duration	~3 hours
You need	A browser + JupyterLab (or Google Colab)
Data	Real metro-train compressor sensors (MetroPT-3, ~144,000 minutes)
Hand in	Your completed notebook AND a written report (the main deliverable)

1. The situation

A metro train's Air Production Unit (APU) — the compressor that feeds the brakes and air suspension — sometimes develops an air leak. When it fails, the train is pulled from service, disrupting passengers and operations. You have four months of real sensor readings from one train's APU, with the genuine air-leak failures marked.

Your job: build a machine-learning model that DETECTS an air-leak from the sensors (a leak that is happening right now), judge how good it really is, and decide how to turn it into a sensible maintenance alert. Note the word detects: this model reads the sensors from the same minute as the failure, so it spots a leak in progress — it is not early warning. This is your first ML lab, so every new idea is introduced one at a time, with a worked example first.

2. What you will be able to do

- Prepare real sensor data as features (X) and a target (y).
- Split time-series data correctly — train on the past, test on the future.
- Train a classifier (Random Forest) and read a confusion matrix.
- Explain why accuracy is the wrong score for rare failures, using precision and recall.
- Choose an alarm threshold by weighing false alarms against missed failures.
- Write a short analytical report that a maintenance manager could act on.

3. What's in your package

File	What it is
lab2_predictive_maintenance.ipynb	The notebook you work in — open this.
metrout3_teaching.csv	The real sensor dataset (already provided).
Student_Report_Lab2.docx	Where you WRITE your report — the main deliverable.
Activity_Dossier_Lab2.pdf	This document — your guide to the activity.
START_HERE.txt	One-page quick start.

4. Before you start

Open JupyterLab (or upload the notebook and the CSV to Google Colab). Keep the notebook and the CSV in the same folder, then run the cells from the top with Shift+Enter. Training the model takes only a few seconds.

```
# In JupyterLab: open a terminal in the folder, then
jupyter lab
# open lab2_predictive_maintenance.ipynb and run each cell top to bottom
```

How the notebook works (about 50/50): each step is worked and explained, then you DO a ☐ Your turn cell yourself (there are eight). It ends with a two-part ☐ Challenge. Only about a third of the ideas are brand-new (features, train/test split, training, metrics, threshold) — and each is introduced gently.

5. The activity, step by step

Step	What you do	Already learned / New?
0–1	Load the sensor data and look at it	Already learned (Lab 1)
2	See how rare failures are	Already learned
3	Plot a sensor around a real failure	Already learned
4	Split into features (X) and target (y)	NEW
5	Split into train / test by time	NEW
6	Train a Random Forest classifier	NEW
7	Judge it: accuracy trap, precision, recall	NEW
8	Set the alarm threshold (the trade-off)	NEW
□	Challenge: key sensors + cheapest threshold	Your choice

6. What to hand in

Deliverable	Notes
Completed notebook	All steps run; the eight □ Your turn cells filled; the □ Challenge attempted.
Written report (main deliverable)	Student_Report_Lab2.docx — your analysis in your own words. See the warning below.
⚠ The report is a report, not a code dump. Do NOT paste code or notebook outputs into it. Use your numbers as evidence, but write in full sentences what they MEAN for the maintenance team. A report that is only pasted outputs scores no marks — the thinking is the point.	

7. How you are assessed (100 marks)

Component	Marks	Full marks means
Notebook: steps run + 8 Your turn cells	30	Notebook runs top to bottom; practice cells completed
□ Challenge (both parts)	15	Key sensors identified; cheapest threshold worked out
Report: analysis & interpretation	40	Explains the findings in your own words (not pasted outputs)
Report: threshold decision + risks	15	A defensible threshold with the trade-off, plus real risks

8. Your data — where it comes from

The dataset is real: MetroPT-3, sensor readings from the Air Production Unit of a Metro do Porto train, 2020. Your file is a 1-minute-averaged teaching version (~144,000 rows); the real air-leak

failures of 18 April and 29–30 May 2020 are preserved. About 1.3% of minutes are failures — realistically rare.

Source: UCI Machine Learning Repository, dataset 791. Licence: Creative Commons Attribution 4.0 (CC BY 4.0) — free to use, including commercially, with attribution. It is provided in your package; you do not need to download anything.

9. Getting unstuck

- Run cells from the top in order — later cells use variables from earlier ones.
- Never split time-series data randomly; train on the past, test on the future.
- A high accuracy score is NOT proof of a good failure detector — check recall and precision.
- Raising the threshold cuts false alarms but starts missing failures — that IS the trade-off.
- Stuck? Re-read the step's Insight box, then ask a teaching assistant.